INCH-POUND
MIL-H-52213F(ME)
29 June 1992
SUPERSEDING
MIL-H-52213E(ME)
17 September 1986

MILITARY SPECIFICATION

HEATERS, HOT OIL

This specification is approved for use within the USA Belvoir Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of' Defense.

1. SCOPE

1.1 Scope. This specification covers three sizes of automatic hot-oil heaters. The heaters shall be skid mounted or trailer mounted, light fuel oil fired, and electric motor driven.

1.2 Classification. The heaters shall be of the following types and sizes, as specified (see 6.2):

Type I - Skid-mounted.
  Size 1 - 1,500,000 Btuh output.
  Size 2 - 2,100,000 Btuh output.
  Size 3 - 3,000,000 Btuh output.

Type II - Trailer-mounted.
  Size 2 - 2,100,000 Btuh output.

2. APPLICABLE DOCUMENTS

2.1 Government documents. .

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3895
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:10Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

FEDERAL

W-C-586	Conduit Outlets Boxes, Bodies, and Entrance Caps, Electrical: Cast Metal.
W-R-550	Rod, Ground (with attachments).
TT-P-664	Primer Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.
UU-T-81	Tags, Shipping and Stock.
VV-F-800	Fuel Oil, Diesel.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-636	Box, Shipping, Fiberboard.
PPP-P-40	Packaging and Packing of Hand Tools.
PPP-T-60	Tape: Packaging, Waterproof.

MILITARY

MIL-C-104	Crate, Wood, Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116	Preservation, Methods of.
MIL-V-173	Varnish, Moisture- and Fungus-Resistant, (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514	Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	Treatment and Painting of Materiel.
MIL-C-717	Castable Mix, Refractory, High-Temperature Hydraulic-Setting.
MIL-F-3541	Fittings, Lubrication.
MIL-C-3774	Crates, Wood; Open 12,000- and 16,000-Pound Capacity.
MIL-G-3859	Grease Guns, Hand-Operated, Lever, Push and Screw Type.
MIL-G-10924	Grease, Automotive and Artillery.
MIL-S-12514	Starters, Motor; Across-the-Line and Reduced Voltage Types, A-C Induction, 3/4 to 100 HP, Manual and Electric.
MIL-P-14105	Paint, Heat-Resisting (for Steel Surfaces).
MIL-B-16008	Brick Insulation, High Temperature, Fire Clay.
MIL-H-18160	Hose Assemblies, Corrugated Steel, 200 Psi and 425 Deg. F Maximum Steam Service Rating.
MIL-I-22023	Insulation Felt, Thermal and Sound Absorbing Felt, Fibrous Glass, Flexible.
MIL-L-22851	Lubricating Oil, Aircraft Piston Engine (Ashless Dispersant).
MIL-G-25013	Grease, Aircraft, Ball and Roller Bearing.
MIL-C-46168	Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-C-52286	Cable Assemblies, Power, Electrical, (With Molded on Terminations).
MIL-C-52286/11	Cable Assembly, Power Electrical.
MIL-C-52286/24	Cable Assembly, Power Electrical, Adapter.
MIL-C-53039	Coating, Aliphatic Polyurethane, Single Component, Chemical Agent Resistant.
MIL-V-62038	Vehicle, Wheeled, Preparation for Shipment and Storage of.
MIL-T-83133	Turbine Fuel, Aviation, Kerosene Type, Grade JP-8.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:10Z
Check the source to verify that this is the current version before use.

STANDARDS

FEDERAL

FED-STD-H28	- Screw Thread Standards for Federal Services.
FED-STD-595	- Colors Used in Government Procurement.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-633	- Mobile Electric Power Engine Generator Standard Family Characteristics.
MIL-STD-642	- Identification Marking of Combat, and Tactical Transport Vehicles.
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines.
MIL-STD-838	- Lubrication of Military Equipment.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.
MIL-STD-1366	- Material Transportation System Dimensional and Weight Constraints, Definition Of.
MIL-STD-1791	- Design for Internal Aerial Delivery in Fixed Wing Aircraft.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1474	- Noise Limits for Military Materiel.
MS52127	- Chock, Wheel.
MS53052	- Bracket Assembly, Liquid Container, Five Gallon.
MS75021	- Connector, Receptacle, Electrical - 12 Contact, Intervehicular, 24 Volt, Waterproof.

HANDBOOKS

MILITARY

MIL-HDBK-113	- Guide for the Selection of Lubricants, Fluids, Preservatives and Specialty Products for use in Ground Equipment Systems.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:10Z
Check the source to verify that this is the current version before use.

DEPARTMENT OF LABOR (DOL)

29 CFR 1910 - Occupational Safety and Health Standards (OSHA)

(Application for copies should be addressed to the local DOL Office or to Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

DEPARTMENT OF TRANSPORTATION (DOT)

Motor Vehicle Safety Standard (FMVSS).
Motor Carrier Safety Regulations (FMCSR).

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

B31.1 - Power Piping.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3953 - Strapping, Flat Steel and Seals.
D 4675 - Selection and Use of Flat Strapping Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN TRUCKING ASSOCIATIONS, INC. (ATA)

Regular Common Carrier Conference (RCCC).
RCCC RP#S110 - Vehicle Running Lamps.

(Application for copies should be addressed to the American Trucking Associations, Inc., 1616 P Street, NW, Washington, DC 20036.)

AMERICAN WELDING SOCIETY (AWS)

B2.1 - Welding Procedure and Performance Qualification.

(Application for copies should be addressed to the American Welding Society, Inc., 550 Northwest Le Jeune Road, Miami, FL 33135.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:10Z
Check the source to verify that this is the current version before use.

ASSOCIATION OF AMERICAN RAILROADS (AAR)

Rules Governing the Loading of Commodities on Open-Top Cars.

(Application for copies should be addressed to the Association of American Railroads, 50 F Street NW, Suite 7702, Washington, DC 20001.)

FRICTION MATERIALS STANDARDS INSTITUTE, INC.

Automotive Data Book.

(Application for copies may be obtained from brake lining supplier or should be addressed to the Friction Materials Standards Institute, Inc., 370 Lexington Avenue, New York, NY 10017.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ICS - 6 - Enclosures for Industrial Controls and Systems.
MG1 - Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 155 East 44th Street, New York, NY 10017.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE HANDBOOK
J10 - Automotive and Off--Highway Air Brake Reservoir Performance and Identification Requirements.
J321 - Tire Guards for Protection of Operator of Earthmoving Halage Machines.
J429 - Mechanical and Material Requirements for Externally Threaded Fasteners.
J431 - Automotive Gray Iron Castings.
J492 - Rivets and Riveting.
J516 - Hydraulic Hose Fittings.
J551 Performance Levels and Methods of Measurement of Electromagnetic Radiation from Vehicles and Devices.
J553 - Circuit Breakers.
J560 - Seven-Conductor Electrical Connector for Truck-Trailer Jumper Cable.
J561 - Electrical Terminals - Eyelet and Spade Type.
J682 - Rear Wheel Splash and Stone Throw Protection.
J683 - Tire Chain Clearance - Trucks, Buses (Except Suburban, Intercity, and Transit Buses), and Combinations of Vehicles.
J694 - Disc Wheel/Hub or Drum Interface Dimensions - Commercial Vehicles.
J697 - Safety Chain of Full Trailers or Converter Dollies.
J712 Industrial and Agricultural Disc Wheels.
J759 - Lighting Identification Code.
J821 - Electrical Wiring Systems for Construction, Agricultural and Off-Road Machines.
J847 - Trailer Tow Bar Eye and Pintle Hook/Coupler Performance.
J849 Connection and Accessory Locations for Towing Multiple Trailers.
J858 - Electrical Terminals Blade Type.
J928 - Electrical Terminals - Pin and Receptacle Type.
J1127 - Battery Cable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:10Z
Check the source to verify that this is the current version before use.

J1128 - Low Tension Primary Cable.
J1234 - Specification Definitions, Off-Road Work Machines.
J1292 - Automotive, Truck, Truck-Tractor, Trailer, and Motor Coach Wiring.
J1402 - Automotive Air Brake Hose and l-lose Assemblies.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT (SCAQMD)

Rule 442 - Usage of Solvents.

(Application for copies should be addressed to the South Coast Air Quality Management District, 9150 E. Flair Drive, El Monte, CA 91731.)

TIRE AND RIM ASSOCIATION, INC. (T&RA)

T&RA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

TRUCK-TRAILER MANUFACTURES ASSOCIATION, INC.

TTMA RP#6
TTMA RP#9
TTMA RP#14
TTMA RP#40
TTMA Bulletin No. 65

(Application for copies should be addressed to the Truck-Trailer Manufacturers Association, 1020 Princess Street, Alexandria, VA 22314.)

UNDERWRITERS LABORATORIES (UL)

UL 6 - Rigid Metal Conduit.
UL 157 - Carbon Dioxide Fire Extinguishers.
UL 514B - Fittings for Conduit and Outlet Boxes.
UL 1242- Immediate Metal Conduit.

(Application for copies should be addressed to Underwriters Laboratory es, Inc., 333 Pfingster Road, Northbrook, IL 60002.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Source: https://assist.dla.mil -- Downloaded: 202640-22T21:10Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Description. Each heater shall consist of an insulated heat exchanger, an oil burner system, an expansion and cold-oil seal tank, a transfer-oil circulating system, controls and safety devices, and all accessories required for operation (except for burner fuel tank). Heaters shall be ready for use after service connections have been made to a user system.

.3.3 Safety. All reciprocating, rotating, or moving parts of equipment shall be guarded when such parts are exposed to contact by personnel or otherwise create a hazard. All platforms, steps, and ladder rungs shall have antiskid surfaces. Heaters shall comply with all applicable FMCSR and FMVSS and Department of Labor Occupational Safety and Health Standards CFR 1910. All hot spots shall be insulated or guarded to prevent exposure of personnel to this hazard.

3.3.1 Radioactive material. Radioactive material shall not be incorporated into the heater. Radioactive material is defined as any material with a specific activity greater than 0.002 microcuries per gram or with an activity exceeding 0.01 microcuries for any part of any component.

3.3.2 Asbestos material. Asbestos material shall not be incorporated into the heater.

3.3.3 Cadmium elating. Cadmium plating or cadmium plated parts shall not be used.

3.4 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.4.1 Material deterioration corrosion control. The heater shall be fabricated from compatible materials that are inherently corrosion resistant or treated in order to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operation or storage environment to which the item may be exposed.

3.4.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.4.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or the designated representative (see 6.2).

3.4.4 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the heater may be newly fabricated from recovered materials to the maximum extent practicable, provided the heater produced meets all other requirements of this specification.

Source: https://assist.dla.mil -- Downloaded: 202640-22T21:10Z
Check the source to verify that this is the current version before use.

Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the heater.

3.4.5 Transfer oil. Unless otherwise specified (see 6.2), transfer oil shall be furnished with each heater. The transfer oil shall conform to MIL-L-22851, type III.

3.4.6 Bearings. Bearings that are subject to heating by hot transfer oil shall be heat stabilized for operation at a temperature of not less than 475 'F and shall be lubricated with grease conforming to MIL-G-25013.

3.4.7 Gaskets and seals. Gaskets and seals that are subject to heating by hot transfer oil shall be oilproofed and shall withstand all temperatures up to 600 "F. Any elastomeric gaskets or seals exposed to hot transfer oil, shall be fabricated from viton or equipment heat-and-oil resistant fluoroelastomer (see 6.2.2).

3.4.8 Refractory. Refractory material shall conform to MIL-C-717 or MIL-B-16008, as applicable.

3.4.9 Pipe and tubing. Unless otherwise specified (see 6.2), pipe and tubing shall be seamless steel conforming to ASME B31.1 for 150 psi, 475 'F hot-oil service.

3.4.10 Valves and pipe fittings. Unless otherwise specified (see 6.2), all valves, caps, couplings, flanges, crosses, elbows, reducers, bends, tees, and nipples shall be made of steel for 150 psi, 475 'F, hot-oil service, or of malleable iron for 300 psi, 475 "F, hot-oil service, as applicable, in accordance with ASME B31.1. Fittings shall be provided with threaded or flanged connections and shall withstand without leakage, permanent deformation, or damage 150 psi, 475 'F, hot-oil service when connected. Threaded fittings shall be National taper pipe thread (NPT). Manually operated valves shall be labeled with double-ended arrows indicating direction of handle movement and labeled at each end to indicate the functional result (i.e., "OPEN", "CLOSE', etc.). The labels shall be inherent to the valves and shall be either sunken or raised from the surface not less than 1/32 of one inch.

3.5 Common parts. Common hardware and fastening devices, such as bolts, nuts, screws, and similar items, shall be held to a minimum number of types and sizes and shall be threaded to the standards of FED-STD-H28.

3.6 Human factors engineering. The heater shall conform to human factors engineering design criteria as described in MIL-STD-1472. Special design emphasis shall be given, but not limited to general requirements (4), control/display integration (5.1), visual displays (5.2), controls (5.4), labeling (5.5), ground workspace design requirements (5.7), design for maintainer (5.9), and hazards and safety (5.13) of MIL-STD-1472, as applicable.

3.7 Noise limits. The noise produced by the heater shall conform with MIL-STD-1474 requirements except 5.2, 5.3, and 5.4, when tested in accordance with 4.5.2.13. Full compliance with the term of MIL-STD-1474, 4.2 and 4.3 shall be accomplished after, and not in lieu of, procedures and documentation accomplished pursuant to MIL-STD-1474, 5.1.1.3.2. The contracting officer is the only official that is authorized to provide the express written permission referenced at

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:10Z
Check the source to verify that this is the current version before use.

5.l.1.3.2.d. The written permission to exceed the specified noise limit(s) shall be obtained before determinations are made regarding the applicability of 4.2 and 4.3 of MIL-STD-1474.

3.8 Performance. The heater, burner, heat exchanger, and main circulating pump shall be capable of performing as follows using both commercial electrical power and electrical power provided from the Government-furnished generator (see 3.30):

a. The maximum heater output shall be not less than 1,500,000 Btuh for size 1, not less than 2,100,000 Btuh for size 2, and not less than 3,000,000 Btuh for size 3.
b. The efficiency of the heat exchanger shall be such that at its rated output the fuel consumption rate of the burner shall be not more than 16, 26, or 32 gallons per hour (gph) for size 1, 2, and 3, respectively.
c* The main circulating pump shall have a capacity of not less than 70, 100, and 140 gallons per minute (gpm) for size 1, 2, and 3 heaters, respectively, at an transfer oil outlet temperature of 450 ‘F and a pump discharge pressure of not less than 50 pounds per square inch gage (psig) for the specified transfer oil (see 3.4.5).
d. The heat exchanger shall maintain a flow rate of not less than 70, 100, and 140 gpm for size 1, 2, and 3, respectively. The heater shall maintain this flow rate at art outlet temperature of 450 +10, -0 “F when the inlet temperature of the transfer oil is not more than 360 “F using the specified transfer oil (see 3.4.5).

3.9 Endurance. When subjected to the tests specified in 4.5.2.6, the heater shall show no evidence of cracking, flaking, shrinking, cement failure, or any other damage.

3.10 Circulating pump motor. Unless otherwise specified herein, the motor shall be 3-phase, 60-cycle, ball-bearing, squirrel-cage-induction class, continuous rating, totally enclosed, fan-cooled, design B, in accordance with NEMA Standard MG1, integral horsepower, class F insulation, temperature rise 95°C. When specified (see 6.2), the motor shall be design C in accordance with NEMA Standard MG1, integral horsepower, class B insulation, temperature rise 40 ‘C. The motor shall be compatible with the 15 kW generator specified in 3.30.

3.10.1 Motor starter. The circulating pump motor shall be provided with a motor starter. The motor starter shall conform to MIL-S-12514, type I, class 2. The starter enclosure shall be in accordance with NEMA ICS - 6, type 3S.

3.10.2 Internal wiring. Internal wiring shall be installed in conduit conforming to UL 6, UL 514B, or UL 1242, as applicable. Conduit outlets shall conform to W-C-586.

3.10.3 Ground rods. Two ground-rod assemblies for grounding the main power supply shall be provided. The ground-rod assembly shall conform to W-R-550, type 111, class B, consisting of a driving bolt, three 3-foot sections of’ 5/8-inch-diameter copper-clad steel rod, and three threaded couplings. A ground wire of not less than 15 feet long shall be provided with each assembly.

3.11 Heat exchanger. The heat exchanger shall be oil-fired; shall be either the fire-tube, liquid-tube, or jacketed type; and shall be of all-welded steel construction.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:10Z
Check the source to verify that this is the current version before use.

3.12 Insulation. The heat exchanger, except the combustion head, oil transmission lines, and overflow lines shall be insulated with either fiberglass felt insulation conforming to MIL-I-22023, type I, class 4, vermiculite solid block insulation, or a ceramic blanket type insulation for temperatures up to 1800 'F. Fiberglass felt insulation, when used, shall be applied to the outside of the welded steel jacket of the heat exchanger. The insulation shall be covered with a waterproof outer shell of not less than 0.0598-inch-thick sheet steel or not less than 0.090-inch-thick sheet aluminum. Vermiculite solid block insulation or ceramic insulation, when used, shall be applied to the inside of the welded steel jacket of the heat exchanger. The insulation shall be capable of maintaining the temperature of the outer shell and rear head at not more than 125 'F, plus the ambient temperature, after the heater has been operated at rated capacity for not less than 4 hours. If the temperature of the outer shell and rear head is more than 85 "F, plus the ambient temperature after the heater has been operated at rated capacity for not less than 4 hours, the outer shell and rear head shall be shielded or guarded as specified in 3.3.

3.13 Burner. The burner shall be a single burner of the air-atomizing, mechanical-atomizing or pressure-atomizing type and shall burn fuel oil conforming to VV-F-800, grades DF-A, DF-1, and DF-2, and JP-8 turbine fuel conforming to MIL-T-83133. The burner shall be equipped with direct electrical ignition and shall be sequenced for automatic operation. The combustion controls shall be of the three-position high-low-off type or of the modulating type. The low firing rate shall be 33-1/3 ±10 percent of the maximum firing rate. The flame-failure reaction time shall be not more than 4 seconds. The burner shall be provided with all pumps, fans, blowers, and other accessories necessary for operation. Two 50-foot, oil-resistant, flexible fuel lines complete with appropriate fittings which can be connected to the fuel pump shall be provided. The fuel pump supply line shall include a replaceable full-flow filtering device. Burner components shall be waterproofed.

3.13.1 Operating sequence. The operating sequence shall be as follows:

Burner start by means of a manual starting switch.

b. Fan and timer start for four air changes to purge the heater. The total purge time shall be not less than 15 seconds.
c. After completion of purge, when all fuel safety and heater interlocks are closed and the combustion controls are at low fire, the timer shall actuate the igniter and open the main fuel safety shutoff valve.
d. When the flame-sensing device does not sense the main flame within 15 seconds after the main fuel valve opens, the device shall lock the burner out to prevent automatic recycling. When this occurs, a manual restart shall be required.
e. After the main flame is detected, the combustion controls shall release from the low-fire position and go to high-fire position, except on the modulating-type burner which shall adjust to heat demand.
f. After satisfying the heat demand, the combustion controls shall cause the burner to go from high-fire position to the off position and shall return sequentially to low-fire position and high-fire position upon a new demand for heat, except for the modulating-type burner which shall adjust to heat demand.
g. Flame failure shall shut down the burner automatically and a manual restart shall be required.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:10Z
Check the source to verify that this is the current version before use.

3.14 Controls and safety devices. Controls and safety devices shall include the following:

a. Indicating temperature regulator of the knob type that can be set manually for the desired transfer oil temperature. The regulator shall be adjustable from 100 to 500 'F. A thermometer with numerical graduations at least every 10 degrees and tic graduations at least every 5 degrees shall be provided to indicate the actual temperature for comparison with the setting. The thermometer shall not be a glass-encased type.
b. High-temperature limit control that shall cut off the burner in the event the regulator fails. This control shall be adjustable from 100 to 475 'F and shall have a positive cutoff at 475 'F.
c. Low-oil-level, microswitch-type, cutoff control that shall cut off the burner when the oil level in the expansion tank falls below 25 percent of capacity level.
d. Fuel-failure safety shutoff control.
e. Flame--sensing device of the ultra-violet-detector type. The main oil heating flame shall be sensed as long as it is present.
f. Flame-failure control of the electronic type. This control shall have a reaction time of not more than 4 seconds.
g. Burner-cutoff control in the event of failure of the main circulating pump.
h. All interlocks and accessories to insure sequence of operation as specified in 3.13.1 and to insure operation of all controls and safety devices.
i. Circuit breaker to protect all live circuits, and a means to manually disconnect the main power supply.

3.14.1 Control box. All automatic controls, manual controls, and temperature-checking devices shall be centrally located in a watertight compartment and shall be accessible to an operator while standing on the ground. Controls shall be in accordance with 3.6. The control box shall have provisions for locking by key-type padlock when not is use. It shall not impede operation of the controls during operation of the heater.

3.15 Transfer oil pumping and piping system.

3.15.1 Main circulating pump. Unless otherwise specified (see 6.2), the main circulating pump shall be of the positive displacement type. The pump shall be provided with antifriction bearings or high-temperature, bushing-type bearings. The main circulating pump shall be provided with a bypass system and relief valve to handle the pump discharge rate. The bypass system shall maintain transfer oil flow through the heat exchanger. All bearings shall have grease fittings which can be lubricated from an external source. Lubrication of bearings with transfer oil will not be permitted. The main circulating pump shall be positioned so that there shall be a positive head of transfer oil contained in a surge tank located above the suction side of the pump.

3.15.1.1 Oil withdrawal from external user system. The pumping system shall be provided with a means for withdrawing the transfer oil from the external user system for storage within the heater.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:10Z
Check the source to verify that this is the current version before use.

3.15.2 Piping system. The piping system shall provide all necessary valves and fittings to interconnect the heat exchanger, pump, and expansion tank and shall terminate with an inlet and outlet service connection. Service connections shall be 2-inch iron pipe size NPT for size 1 heater, 2-1/2 inch iron pipe size NPT for size 2 heater, 3-inch iron pipe size NPT for size 3 heater, and shall be equipped with caps and cap-retaining chains or terminal valves.

3.15.2.1 Strainer. The transfer oil line shall include a full-flow oil strainer. The transfer oil line shall include a strainer bypass system to permit full-flow o-ii flow for operation during a strainer element cleaning. The strainer and strainer accessories shall be reusable. The strainer body and the strainer elements shall be made of corrosion-resistant material.

3.15.2.2 Connections. All threaded connections shall be NPT. With the exception of connections that are necessary for removal of components, all threaded connections shall be externally brazed, silver soldered, or welded to prevent leakage. Flanged connections with gaskets shall be used to facilitate component removal.

3.16 Expansion tank. A steel expansion tank shall be provided. The tank capacity shall provide space for the expansion of transfer oil being heated in the heat exchanger and being circulated to the user system, a space for transfer oil storage, a drain, an overflow, an air vent valve, a provision for filling, a provision for visually determining the oil level, all necessary controls (see 3.14), and all other accessories necessary for heater operation as specified herein. The expansion tank may be used as the cold-oil seal specified in 3.16.1. The expansion tank need not be insulated but shall be isolated from the heat exchanger and located at an elevation above the pump. The minimum capacity of the expansion tank shall be twice the volume required by the expansion of the transfer oil in a fully-charged system when the temperature of the transfer oil is raised from 70 to 475 ‘F. Transfer oil in a fully-charged system shall be that amount of oil required to fully charge the heat exchanger and surge tank (see 3.15.1) plus not less than 180 gallons for the external user system. The expansion tank, heat exchanger, and surge tank shall store all the transfer oil.

3.16.1 Cold-oil seal. A cold-oil seal shall be provided to protect the transfer oil against oxidation. A separate uninsulated tank may be used as the cold-oil seal in combination with the expansion tank, and shall be isolated and located as specified in 3.16. The cold-oil seal shall have a vent at the top to permit air to escape and shall incorporate a provision for visually determining the oil level, a drain, and a 50 to 500 “F thermometer. After not less than 8 hours of operation of the heater at rated output, the temperature of the transfer oil exposed or in contact with the atmosphere shall not exceed the ambient temperature +75 “F, when measured within 2 inches of the oil surface.

3.17 Type I. frame-skid-mounted. The components of the type I heater shall be supported on a welded structural steel framework consisting of angles, channels, pipe, or I-beams, reinforced with crossmembers. l The framework shall be bolted to a removable skid-type base, or shall be fabricated integral with a skid-type base. The main skid members shall be curved or turned up at both ends for towing, and shall be provided with two towing eyes. The skids and skid fastenings shall withstand towing without damage or permanent deformation when the heater is towed on dry level surfaces at speeds of not less than 2 mph or more than 5 mph.

Source: https://assist.dla.mil -- Downloaded: 2026l0-22T21:10Z
Check the source to verify that this is the currentversion before use.

3.18 Type II. trailer-mounted. All components of the heater, including the generator (see 3.18.2), shall be mounted on a pneumatic-tired trailer. The prime mover of this trailer shall be a 5 ton series truck. 'The heater configuration shall be either an integral frame type for both heater and vehicular components or a separate frame for the heater installed on a trailer chassis assembly. Except as specified herein, the heater trailer components shall be in accordance with the following, (in order of precedence):

a. Department of Transportation (DOT) Federal Motor Safety Standards (FMVSS).
b. DOT Motor Carrier Safety Regulations (FMCSR).
c. Truck-Trailer Manufacturers Association (TTMA).

3.18.1 Trailer.

3.18.1.1 Weights and dimensions. The gross vehicle weight (GVW) of the heater shall include the weight of the heater components accessories, attachments, heat transfer oil, lubricants, and vehicular components. Vehicular component ratings shall not be raised to conform to the requirements of this specification. With the heater level on level ground and with the tire inflation pressure as recommended in the T&RA Yearbook and Supplementary Service Data for the imposed GVW loading and highway speed, the physical characteristics of the heater shall be as follows:

	GVW, maximum	12,000 pounds.
b.	Vertical lunette load, range	5 to 10 percent of GVW.
	Overall length, maximum	240 inches.
d.	Overall width, maximum	96 inches.
e.	Overall height, maximum (in travel condition with stack removed)	102 inches.
f.	Ground clearance, minimum (excluding running gear)	12 inches.
g.	Lunette height range	31 to 37 inches.
h.	Angle of departure, minimum (SAE J1234)	20 degrees.
i.	Suspension limits	
	(1) GVW to 8,000 pounds, maximum	single axle.
	(2) GVW to 12,000 pounds, maximum	tandem axle.

3.18.1.2 Mobility. The heater at GVW shall perform as specified herein when towed over the roads and terrain specified, by a truck matched to the heater GVW. The heater shall operate over smooth, hard-surfaced primary highways at sustained speeds up to 45 miles per hour (mph) and over gravel and unimproved roads at speeds up to 25 mph.

3.18.1.2.1 Trailing and turning ability. The heater shall trail the towing truck without cramping, weaving, side-slipping, or bottoming. The heater shall follow the truck without exceeding the tracking deviation as specified in DOT Motor Carrier Safety Regulation 393.70, when towed over a paved surface. Turns of at least 60 degrees, as measured from straight ahead to the right and to the left shall not cause damage. or interference between the heater and the truck.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:10Z
Check the source to verify that this is the current version before use.

3.18.1.2.2 Braking. When tested as specified in 4.5.2.2.1, the combined service brakes of the truck and the heater shall control, decelerate, and stop the vehicle combination from a speed not less than 20 MPH within 30 feet and be in accordance with DOT Motor Carrier Safety Regulation 393.52.

3.18.1.3 Intervehicular connections. The heater shall be furnished with an intervehicular electrical harness and two airbrake lines for connecting the towing vehicle electrical and airbrake systems to the heater systems. All intervehicular connections shall be installed on the heater so that they extend between 36 and 42 inches toward the towing vehicle from the center of the lunette eye. Two fixed dummy couplings shall be installed to hold the airbrake line ends with the lines doubled back. Intervehicular connections, when connected and when stowed, shall be protected. When a 7 and 12 pin receptacle is specified (see 3.18.1.13.1), intervehicular harnesses compatible with each receptacle shall be provided as specified herein.

3.18.1.4 Drawbar components. Each heater shall be equipped with a towing attachment having a lunette conforming to SAE J847 installed in accordance with SAE J849. The lunette shall be installed on the towing attachment within l/8-inch on either side of the centerline of the heater. The towing attachment shall be of the fixed type, constructed and installed to provide a structure necessary to resist pulling, bending, and twisting without permanent deformation or damage under all transport conditions specified herein. Two safety chains conforming to SAE J697, with two safety hooks, shall be attached to the trailer frame and extend between 36 and 42 inches forward of the center of the lunette eye.

3.18.1.5 Support legs. At least three support legs shall be provided. A means shall be provided for lifting the vehicle at its maximum GVW into a position where the maximum GVW less the tires and suspension system is supported by the lowered support legs. The support legs shall be capable of leveling the heater at any position up to and including a 10 degree slope. The force provided by the operator to lift the unit onto the support legs and to level the unit shall not exceed the requirements of MIL-STD-1472, design for maintainer (5.9) as per 3.6. The support legs shall have bearing pads. The edges of the bearing pads shall be turned up. The landing pads shall have a ground bearing pressure of not more than 20 pounds per square inch when supporting the heater maximum GVW. The up turned portion of the bearing pads shall not be figured into the ground bearing pressure calculations.

3.18.1.6 Frame. The frame shall be of sufficient strength and rigidity to support and connect all components, maintain alignment, and distribute the stress of the suspension mounts in order to meet the transport and operating conditions specified herein.

3.18.1.6.1 Rear bumper. A bumper not less than 8 inches in section depth shall be provided at the rear of the heater. The bumper shall extend rearward at least 2 inches beyond any component of the heater. The bumper shall extend the full width of the heater and the ends shall be rounded or wrap-around to the extent necessary to prevent hooking when sideswiped. The bumper shall conform to the safety requirements of DOT FMCSR.

3.18.1.7 Suspension. The heater suspension system shall withstand static and dynamic loads, when operated as specified herein, without permanent deformation, damage, or misalignment. The suspension system, in conjunction with the tires,

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:10Z
Check the source to verify that this is the current version before use.

shall protect the heater from damaging shock and vibration. The suspension system shall have a rated capacity at least equal to the load imposed on each member, measured at the ground. The suspension system shall be of the nonlubricated type. The suspension system shall provide a GVW distribution as specified in 3.18.1.1. The heater suspension shall have sufficient articulation to permit tandem axle heaters to roll over 6-inch-high, square obstacles with any wheel without loss of tire-to-surface contact of the remaining wheels. Each axle and trunnion mounting shall be cushioned with flexible, elastomer material. Elastic stops or bumpers shall prevent metal-to-metal contact at maximum jounce and rebound position of all wheels. The axle(s) of the heater shall be perpendicular to and symmetrical about the heater's longitudinal centerline. The difference between the diagonal measurements from the center of the lunette eye on the heater centerline to the center end of each front spindle. or tandem trunnion shall not exceed 1/16 inch. The alignment of each subsequent axle shall result in these components being parallel to the forward axle spindles or tandem trunnion. The difference between all corresponding spindle centers on each side shall not exceed 1/8 inch. Tire and wheel assemblies for all axle combinations shall be balanced within industry limits to eliminate wheel tramp and drumming at road speeds up to 40 mph. On single axle heaters, two shock absorbers shall be provided to control jounce and rebound. The two shock absorbers shall be mounted to the frame and to the axle on each side of the heater with elastomer bushings.

3.18.1.8 Single and tandem axles. The axle(s) shall be rated by the axle manufacturer at a continuous load capacity rating of not less than 150 percent of the heater weight when loaded to maximum GVW. The spindles shall be alloy steel, hardened and ground, and shall be in the correct alignment position with the axle tube. The spindles shall be equipped with tapered roller bearings with cups, bearings, and cones in matched sets from the same bearing manufacturer.

3.18.1.9 Wheels, rims. and tires. The heater shall be equipped with heavy duty type, disc wheel and tire assemblies conforming to TR&A Yearbook and requirements as specified herein. All wheels shall conform to SAE J694 or J712. All tires shall be radial ply with rating of not less than load range F. All wheels and tires furnished shall be identical. A spare tire and wheel assembly shall be provided and mounted on each heater. Tires shall be not more than one year old from the date of production to the date of acceptance by the Government.

3.18.1.9.1 Balancing. Tires, wheels, hubs, and brakedrums shall be inherently in-balance, singly and collectively after assembly, to preclude wheel tramp, drumming, or excessive tire wear, when operated up to the maximum highway speeds specified herein.

3.18.1.10 Fenders and splash guards. The heater shall be equipped with fenders and rear splash guards. The fenders and splash guards shall be in accordance with SAE J321, SAE J682, and SAE J683. The fenders shall support a weight of not less than 200 pounds imposed on any one square foot area. The splash guards shall be not less than 12 by 18 inches with antisail characteristics.

3.18.1.11 Brake system. The heater shall be equipped with a waterproof, service and emergency breakaway brake system, actuating foundation brakes on all wheels. The brake system shall comply with DOT requirements as specified in 3.18, except as specified herein, and shall perform as specified in 3.18.1.2.2 under all conditions. All components and lines of the brake system shall be located, installed, and protected to prevent damage or malfunction from water, ground

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:10Z
Check the source to verify that this is the current version before use.

obstructions, and debris and to obtain maximum efficiency and ease of maintenance. The brake system shall be either straight air type or air-over-hydraulic type, as specified herein. Either brake system shall include (1) service and emergency lines as specified in 3.18.1.11.2; (2) service and emergency air line filters or air cleaners; (3) SAE J10 air reservoir with protected drain cock; and (4) emergency relay valve.

3.18.1.11.1 Air-over-hydraulic brakes. The air reservoir should have a volume capacity of not less than 900 cubic inches for single-axle heaters and not less than 1,750 cubic inches for tandem-axle heaters. One power cluster should be furnished on single-axle heaters. Two power clusters should be furnished on tandem-axle heaters.

3.18.1.11.2 Brake lines. Intervehicular air lines should be furnished with connectors equipped with metal spring covers, identification tags, and coil spring hose guards on each end. The airbrake lines should conform to SAE J1402, type D, class II, and SAE J516, 100R5 reusable fittings.

3.18.1.11.3 Brakedrums. The brakedrums shall be of the anti-bellmouth type, cast and machined from automotive gray cast iron conforming to SAE J431, grade G3500 or G4000.

3.18.1.11.4 Brake linings. The brake linings shall be of the nonfading-from-heat and -from-water-immersion type. The lining shall be of a standard size as listed in the Automotive Data Book of the Friction Materials Standards Institute, Inc.

3.18.1.12 Chock blocks and cradles. The heater shall be provided with two chock blocks and two cradles. One chock block and cradle shall be installed on each side of the heater so that the block may be positioned either to the front or to the rear of the wheel assembly with the captive fastener attached to the heater. The chock blocks shall conform to MS52127, type A or C. The cradles shall securely hold the chock blocks during transport.

3.18.1.13 Vehicular electrical system. Unless otherwise specified herein, the heater shall be equipped with a waterproof, 12-volt vehicular electrical system. Except as specified herein, the electrical system shall be in accordance with the following (in order of precedence):

a. DOT Motor Vehicle Safety Standards (FMVSS).
b. DOT Motor Carrier Safety Regulations (FMCSR).
SAE Handbook, Electrical Equipment and Lighting Section.
d. TTMA Recommended Practices RP No. 9, RP No. 40, RP No. 6, and Bulletin No. 65.

All components shall be provided with the maximum practical protection from damage, shock, and vibration through means of location and resilient mounts. The electrical components shall be in accordance with table L Item quantities that are not specified shall be furnished as required (AR) for each heater.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:10Z
Check the source to verify that this is the current version before use.

TABLE I. Vehicular electrical system.

Description	Requirements	Quantity Per Heater
7 Contact receptacle	SAE J560 and TTMA RP No. 40, 7-wire type with plated metal connectors and waterproof, spring loaded metal cover.	1 ea
Connection box	Waterproof type with either stud and nut terminals or spring loaded terminals.	1 ea min.
Circuit breakers	SAE J553 protecting all live circuits.	AR
Wiring	SAE J1127 or SAE J1128, heavy duty type with 6 rubber or thermosetting insulation only. Wire size SAE J1292, min. 14 gage.	AR
Wire identification	SAE J821	AR
Wiring harness	Polyethylene electrical conduit, each waterproofed with silicone sealant or with one-half overlapping turns of plastic electrical tape.	AR
Grommets	Split type where needed along the wire routing to prevent chafing.	AR
Supports	Elastomer cushioned support straps or elastomer ties with mounts; wiring support spacing max 36-in. span, prevent whipping, chafing, and fatigue damage.	AR
Wiring splices	Mechanical splicing devices; interior waterproofed with silicone lube or compound.	AR
Connectors and terminal	Waterproof types; blade type SAE J858, pin type SAE J928, and spade SAE J561. Blade and pin types fully enclosed and lubricated, prior to assembly with silicone lubricant; spade type equipped with heat shrinkable sleeves and coated with electrical varnish after assembly; all connectors and terminals plated or solder-coated to retard corrosion.	AR
Connector holder clip	Plastic or plastic-insulated metal spring clips; fastened to vehicle; furnished to anchor wire-to-wire friction connectors.	1 ea. per friction connector

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:10Z
Check the source to verify that this is the current version before use.

TABLE I. Vehicular electrical system. (continued)

Description	Requirements	Quantity Per Heater
Running lamps	4 side-marker, 4 clearance, 3 rear identification; per RCCC RP No. S110/1 or RP No. S110/2, flush lamp or low-profile lamp; waterproof and shock resistant types.	11 ea. min.
Combination lamps	2 turn and tail outboard and 2 stop and tail inboard; Per TTMA RP No. 14 and SAE J759, SAE-IST; flush, elastomer housing with bulb and sealing snap-on, or gasketed lens, or sealed unit with elastomer ring; round 4-in. diameter size installed in 4-1/2-in. diameter hole; double contact; 3 wire type including ground wire; waterproof and shock resistant type; installed in the rear bumper.	4 ea. min.
Lamp lenses and bulbs	Polycarbonate plastic type lenses; heavy duty or fleet service type bulbs; bulbs mounted in brass vibration-protected sockets with ceramic or plastic insulators and stainless steel or rubber bulb tension devices.	AR
Reflectors	2 on each corner, round type, securely fastened by bolts, screws or rivets.	8 ea. min.
Wiring diagram	Plate furnished on trailer per TTMA Bulletin No. 65.	1 ea.

3.18.1.13.1 Conversion to 24 volt electrical system. Unless otherwise specified (see 6.2), the heater shall be furnished with a 12-contact receptacle conforming to MS75021-1 in addition to the 7-contact receptacle. The 12-contact receptacle shall be provided with resistance to each circuit to reduce the voltage from the military designed towing vehicle. The 24-volt, 12-contact receptacle shall be connected through the necessary resistors into the following 12-volt trailer circuits as follows (letters depict the 24 volt contact terminals and the numbers refer to the SAE J560 receptacle):

A - Do not connect.
B - Connect to circuit No. 3.
C - Do not connect.
D - Connect to circuit No. 1.
E - Connect to circuits No. 2 and 6.
F - Do not connect.
H - Do not connect.
J - Connect to circuit No. 5.

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:10Z
Check the source to verify that this is the current version before use.

K - Connect to circuit No. 7.
L - Do not connect.
M - Do not connect.
N - Do not connect.

Circuits B and J of the 12-contact receptacle on tactical trucks are combination stop and turn indicator circuits. The normal 12-volt signal lights will function both as turn signals and stop lights and the normal 12-volt stop lights will not be in operation when the trailer is connected to a towing vehicle with a 24-volt power supply. Because of this condition the No. 4 on stop light circuit is not connected to the 24-volt, 12-contact receptacle. Circuit breakers in accordance with SAE J553 shall be furnished. Each receptacle (7- and 12-contact) shall be installed in its own weatherproof connection box, and shall have a spring-loaded cover. Each receptacle shall be permanently identified as to the applicable voltage.

3.18.2 Power supply. Power to operate the heater shall be supplied by a Government-furnished 15 kW, 60 Hz, Mobile Electric Power (MEP) generator set (see 3.30).

3.18.2.1 Power cable and reel. The main feeder cable assembly shall conform to MIL-C-52286 and MIL-C-52286/11 except that the cable shall be type CO-03HLF(3/4-3/10R)1170. A manually-operated, steel reel to contain the 100-foot cable shall be permanently mounted on the heater. Provision shall be made to secure the cable on the reel.

3.18.2.2 Power cable adapter assembly. A power cable adapter assembly shall be provided. The cable assembly shall conform to MIL-C-52286 and MIL-C-52286/24 except the length of the cable shall be 5 feet ±6 Inches, and the cable shall be type CO-03HLF(3/4-3/10R)1170.

3.18.2.3. Main power supply connectors. One each main power supply connector shall be provided conforming to MIL-C-52286, figures 29 and 43.

3.18.3 Flexible hose. Flexible hose shall conform to MIL-H-18160, class S, 2-inch inside diameter, 8-foot lengths, except protective wire guard and couplings shall not be furnished. The flexible hose shall be capable of hot-oil service at temperatures up to 475 “F. Four 8-foot lengths of the specified flexible hose and four 2-inch, all-steel unions with stainless steel seats shall be provided to enable connection of the flexible hoses to one another and to the inlet and outlet service connections.

3.18.3.1 Container for flexible hose. A storage container shall be provided to accommodate the four 8-foot lengths of flexible hose and four 2-inch steel unions specified in 3.18.3. The container shall be fabricated as specified in 3.20.

3.19 Fire extinguisher. The heater shall be provided with a mounted, 10-pound carbon dioxide, Coast Guard approved fire extinguisher and mounting bracket conforming to UL 154.

3.20 Toolbox. A toolbox shall be provided and shall be of a size to hold maintenance tools and ground rod assemblies (see 3.10.3). The toolbox shall be made of steel 0.074-inch nominal thickness and shall include provisions for drainage in its mounted position. The toolbox shall have a hinged lid. One side

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:10Z
Check the source to verify that this is the current version before use.

shall be provided with not less than two fasteners that will keep the lid closed. The toolbox lid shall have an overlapping l-inch turned down lip to prevent entry of rain or dirt. The toolbox shall be securely mounted in a protected and accessible location permitting the lid to be opened through an arc of 90 degrees. The toolbox shall have provisions for locking by key-type padlock.

3.21 Decontaminating apparatus, portable (DAP), M13. The hot oil heater DAP mounting bracket (MS53052-1) shall be installed in a manner which shall not interfere with the normal operation of the equipment. An upright installation is preferred. Horizontal mounting shall be used only with the approval of the contracting officer. The DAP device has a flash point of 165 “F and shall be installed away from exhaust or heat sources.

3.22 Rain. When subjected to the tests specified in 4.5.2.9, the heater shall show no water leakage into the engine exhaust system, burner system, burner exhaust, control box, exposed safety components, electrical conduits, or junction boxes.

3.23 Transportability. The heater shall be capable of being transported (see 6.2), by military or commercial trailers, trains, marine vessels, aircraft and must be capable of withstanding the impact forces encountered in shipment without damage or permanent deformation. The heater shall be equipped with tiedown and slinging provisions.

3.23.1 Tiedown provisions. The tiedown provisions shall conform to MIL-STD-209, class 2 or 3, type II equipment, and to MIL-STD-1791 for equipment restraint criteria. The tiedown provisions shall satisfactorily complete the pull testing as specified in 4.5.2.10, without weld failure, permanent deformation, cracking, loosening, or breaking of the provision or its connecting structural components. Tiedown points shall be labeled “TIEDOWN HERE” and combination lift/tiedown points shall be labeled “LIFT/TIEDOWN HERE”. The labels shall conform to MIL-STD-1472.

3.23.2 Slinging provisions. The slinging provisions shall conform to MIL-STD-209, class 1 or 3, type II equipment. The provisions shall enable the complete heater to be lifted in the normal operating position. ‘ The provisions shall be located so that not less than l-inch clearance is maintained between slings and all exterior parts and shall be fastened to members which will withstand stresses in the amount and direction of pull specified for the provisions without weld failure, permanent deformation, cracking, loosening, or breaking of the provision or its connecting structural components. Slinging provisions may also be used as tiedown provisions when such provisions meet the requirements specified in 3.23.1. Lift points shall be labeled “LIFT HERE”. The labels shall conform to the requirements of MIL-STD-1472.

3.23.3 Air transportability. The heater shall meet the requirements of MIL-STD-1791 for air transport. When required (see 6.2), the contractor shall provide sufficient information (dimensions and weight) to the Government to confirm that the heater being supplied can be loaded and transported by the C-130, C-141, and C-5A aircraft.

3.23.4 Rail transportability. The heater shall be rail transportable in CONUS and NATO countries without restrictions. When loaded on a 50-inch (127 cm) high rail car, the heater shall have a dimensional profile within the AAR outline

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:10Z
Check the source to verify that this is the current version before use.

diagram and the Gabarit International de Chargement (GIC) or PPI, as specified in MIL-STD-1366. The heater shall be capable of withstanding shock loads resulting from rail impact testing in accordance with 4.5.10.4 without failure, damage, or permanent deformation.

3.23.5 Highway transportability. When loaded on a semitrailer/tractor, the heater shall be within the highway permit limits of all states.

3.23.6 Marine transportability. The heater shall be transportable by LARC-V and larger vessels operating over any type of waterway. The heater shall withstand without damage, the shock, rolling, and pitching normally experienced in marine transportation on the deck or in the hold of a cargo vessel. For design purposes, the following conditions are assumed for marine transportation:

Normal roll shall not exceed 15 degrees.
b. Normal pitch shall not exceed 10 degrees.

3.24 Electromagnetic interference. The electromagnetic interference emission characteristics of the heater shall conform to SAE J551.

3.25 Lubrication. All surfaces that require lubrication shall be provided with a means for lubricating.

3.25.1 Lubricants. The procedure for the selection of lubricants shall be in accordance with MIL-STD-838, section 5. Lubricants selected shall be in accordance with MIL-HDBK-113, chapter 2.

@ 3.252 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541, type I, H, or 111. Fittings shall be located in a protected position and shall be accessible to a hand-operated grease gun conforming to MIL-G-3859 equipped with a IO-inch flexible extension. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Where holes are cut in covers, plates, etc., to provide accessibility to fittings, spring-loaded covers shall be provided. Panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility. ,

3.25.3 Pressure-release device. An automatic pressure-release device shall be provided where pressure lubricating equipment could damage grease seals or other components.

3.25.4 Filling, draining, and checking provisions. Enclosures such as gear cases and transmission housings which contain a reservoir of lubricants for the lubrication of the parts enclosed shall be equipped with dipsticks, or sight gages to determine the level of the lubricant. Each enclosure shall be equipped with a means for filling the enclosure with lubricant and for draining. The drain shall be fitted with a magnetic drain plug. The drain outlet shall be located so that removal of the plug will result in complete drainage of the lubricant from the enclosure when the heater is in its normal position. Integral tubes or troughs may be used to convey the lubricant from the drain outlet to a container. Accessibility to the drain plug, the filling means, and the lubricant level checking device shall be provided. All drains shall be labeled to indicate the item being drained (i.e., “EXPANSION TANK DRAIN”, etc.).

Source: https://assist.dla.mil -- Downloaded: 2026l0-22T21:10Z
Check the source to verify that this is the current version before use.

3.25.5 Grease lubrication. All grease lubrication, except for the high temperature grease specified in 3.4.6, shall be with greases conforming to MIL-G-10924. The heater shall be assembled, run-in, tested, and delivered with these greases. The bearings shall be cleaned before lubrication, because military greases are not always compatible with other greases. Each component shall be tagged in a conspicuous place to indicate which military grease has been used.

3.26 Fungus- and moisture-resistance. The electrical circuitry, including all components and connections, except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, composition I or 11, as applicable (see 3.26.1), with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish:

a. Components or circuit elements that are inherently fungus- and moisture-resistant or which are hermetically sealed need not be treated.
b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.26.1 Composition. Composition II shall be used only in the case when local air pollution regulations governing the application of varnish precludes the use of composition I. When composition II is used, the contractor shall provide evidence to the Government that the use of composition II is required, and shall certify that the composition II material complies with Rule 442 of the South Coast Air Quality Management District.

3.27 Treatment and painting. The portions of the heater normally painted, both interior and exterior, shall be cleaned, treated, and painted in accordance with MIL-T-704, type F or G, as applicable. Unless otherwise specified (see 6.2), exterior topcoat color shall be camouflage green 383. The identification plate (see 3.29) shall state “PAINTED WITH CARC”. Heat resistant paint conforming to MIL-P-14105 in accordance with FED-STD-595 color number 34094 shall be used on surfaces exceeding 400 ‘F. When specified (see 6.2), the heater shall be overcoated in accordance with the Government furnished camouflage patterns using either MIL-C-46168 or MIL-C-53039 topcoat or MIL-P-14105 for surfaces which exceed 400 “F in the colors specified in the camouflage patterns. The first article need not be overcoated in camouflage patterns.

3.28 Camouflage pattern data. When specified (see 6.2), the contractor shall provide camouflage pattern data.

3.29 Identification marking/labels. The heater shall be identified in accordance with MIL-STD-130. All marking/labels shall be applied to the heater on plates conforming to MIL-P-514, type I, style 1, composition C of type I, grade A, class 1 material. Plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:10Z
Check the source to verify that this is the current version before use.

3.29.1 Shipping data plate. A shipping data plate conforming to MIL-P-514, type 111, composition C, of type I, grade A, class 1 material shall be provided. In addition, the plate shall show the silhouette of the heater in transport position indicating the center of gravity and the location and capacity of the lifting and tiedown provisions. The plates shall be attached by screws, bolts, or rivets in a location where the plate shall be both visible and legible.

3.29.2 Instruction plates. Each heater shall be quipped with instruction plates or diagrams, Including warnings and cautions, describing any special or important procedures to be followed in assembling, operating, or servicing the heater. Instruction plates shall conform to MIL-P-514, type III, composition C, of type I, grade A, class 1 material. Plates shall be attached by screws, bolts, or rivets in a location where the plate shall be both visible and legible.

3.29.3 Warnings. Warnings or cautions shall be provided for all inherent dangers which can stem from unsafe operation or an unsafe operating position. Warnings or cautions shall be stenciled in Green 383 paint over black, and stenciled in black paint over Green 383 and Brown 383 paint.

3.29.4 Stenciling. The gross weight of the heater shall be stenciled on each side of the heater. On type II heaters, the T&RA yearbook prescribed tire pressure shall be stenciled TP (specify pounds) on each fender. All stenciling shall be in block-type letters not less than 2 inches high or higher than (Army 4 inches maximum) as specified in MIL-STD-642, 5.1.2. Registration numbers shall be marked on the heater in accordance with MIL-STD-642. Color shall be as specified in MIL-T-704.

3.30 Government-furnished property. Unless otherwise specified (see 6.2), the following property will be furnished by the Government (see 6.5):

Item	Description	Identification	Quantity
1	15 kW generator	MIL-STD-633 NSN 6115-00-118-1241	1
2	Camouflage paint pattern		As required

3.31 Workmanship The heater shall be free from defects such as incomplete welds, rust, cracks, and other defects that could impair the operation of the' heater. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges, and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.31.1 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping, wire brushing, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading. Parent materials, weld filler metals, and fabrication

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:10Z
Check the source to verify that this is the current version before use.

techniques shall be as required to enable the heater to conform to the examination and test requirements specified in section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 3/16 inch, unless appropriate bridging techniques are used. Unless otherwise specified, the welding process used in fabrication of the heater shall be at the option of the contractor (see 6.2).

3.31.1.1 Welder and welding operators certification. Welders and welding operators shall be certified according to AWS 62.1. Certification shall be made available for review by the contracting officer or designated representative.

3.31.2 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. Washers or lockwashers shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement.

3.31.3 Riveted connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member, and shall be in accordance with SAE J492.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. Inspections shall be classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c* Inspection of packaging (see 4.6).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:10Z
Check the source to verify that this is the current version before use.

4.3 First article inspection.

4.3.1 Examination. The first article shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article shall be tested as specified in 4.5.2.1 through 4.5.2.13. All first article tests shall be run using JP-8 fuel conforming to MIL-T-83133, grade JP-8. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection. Quality conformance tests shall be run using either JP-8 fuel conforming to MIL-T-83133 or diesel fuel conforming to VV-F-800.

4.4.1 Examination. Each heater shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests. Each heater shall be tested as specified in 4.5.2.7, except that the heater shall be restarted only one time in lieu of 25 times. Failure of the test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The heater shall be examined for the following defects:

101 l Heater not as specified (see 3.2).
102. Safety not as specified (see 3.3).
103. Radioactive material incorporated into equipment (see 3.3.1).
104. Asbestos material incorporated into equipment (see 3.3.2).
105. Cadmium plating used (see 3.3.3).
106. Material not as specified (see 3.4).
107. Material deterioration and control not as specified (see 3.4.1).
108. Dissimilar metals not as specified (see 3.4.2).
109. Identification of materials and finishes not as specified (see 3.4.3).
110. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the heater (see 3.4.4).
111. Transfer oil not as specified (see 3.4.5).
112. Bearings not as specified (see 3.4.6).
113. Gaskets and seals not as specified (see 3.4.7).
114. Refractory material not as specified (see 3.4.8).
115. Pipe and tubing not as specified (see 3.4.12).
116. Valves and pipe fittings not as specified (see 3.4.13).
117. Common parts not as specified (see 3.5).
118. Human factors engineering not as specified (see 3.6).
119. Noise limit not as specified (see 3.7).
120. Circulating pump motor not as specified (see 3.10).
121. Motor starter not as specified (see 3.10.1).
122. Internal wiring not as specified (see 3.10.2).
123. Ground rods not as specified (see 3.10.3).
124. Heat exchanger not as specified (see 3.11).
125. Insulation not as specified (see 3.12).
126. Burner not as specified (see 3.13).
127. Controls -and safety devices not as specified (see 3.14).
128. Control box not as specified (see 3.14.1).
129. Transfer oil pumping and piping system not as specified (see 3.15.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:10Z
Check the source to verify that this is the current version before use.

130. 0il withdrawal system not as specified (see 3.15.1.1).
131. Piping system not as specified (see 3.15.2).
132. Strainer not as specified (see 3.15.2.1).
133. Connections not as specified (see 3.15.2.2).
134 l Expansion tank not as specified (see 3.16).
135. Cold-oil seal not as specified (see 3.16.1).
136. Type I heater not frame-skid-mounted as specified (see 3.17).
137. Type II heater not trailer-mounted as specified (see 3.18).
138. Trailer not as specified (see 3.18.1).
139. Weights and dimensions not as specified (see 3.18.1.1).
140. Mobility not as specified (see 3.18.l.2).
141. Trailing and turning ability not as specified (see 3.18.l.2.1).
142. Braking not as specified (see 3.18.l.2.2).
143. Intervehicular connection not as specified (see 3.18.1.3).
144. Drawbar components not as specified (see 3.18.1.4).
145. Support legs not as specified (see 3. 18.1.5).
146. Frame not as specified (see 3.18.1.6).
147. Rear bumper not as specified (see 3.18.1.6.1).
148. Suspension not as specified (see 3.18.l.7).
149. Single and tandem axles not as specified (see 3.18.1.8).
150. Wheels, rims, and tires not as specified (see 3.18.1.9).
151. Balancing not as specified (see 3.18.1.9.1).
152. Fenders and mudguards not as specified (see 3.18.l.10).
153. Brake system not as specified (see 3.18.1.11).
154. Air-over-hydraulic brakes not as specified (see 3.18.1.11.1).
155. Brake lines not as specified (see 3.18.1.11.2).
156. Brakedrums not as specified (see 3.18.1.11.3).
157. Brake linings not as specified (see 3.18.1.11.4).
158. Chock blocks and cradles not as specified (see 3.18.1.12).
159. Vehicular electrical system not as specified (see 3.18.1.13).
160. Conversion to 24 volt electrical system not as specified (see 3.18.1.13.1).
161. Power cable and reel not as specified (see 3.18.2.1).
162. Power cable adapter assembly not as specified (see 3.18.2.2).
163. Main power supply connectors not as specified (see 3.18.2.3).
164. Flexible hose not as specified (see 3.18.3).
165. Container for flexible hose not as specified (see 3.18.3.1).
166. Fire extinguisher not as specified (see 3.19).
167. Toolbox not as specified (see 3.20).
168. Decontaminating apparatus not as specified (see 3.21).
169. Tiedown provisions not as specified (see 3.23.1).
170. Slinging provisions not as specified (see 3.23.2).
171. Air transportability not as specified (see 3.23.3)
172. Rail transportability not as specified (see 3.23.4).
173. Highway transportability not as specified (see 3.23.5).
174. Marine transportability not as specified (see 3.23.6).
175. Lubrication not as specified (see 3.25).
176. lubricants not as specified (see 3.25.1).
177. Lubrication fittings not as specified (see 3.25.2).
178. Pressure-release device not as specified (see 3.25.3).
179. Filling, draining, and checking provisions not as specified (see 3.25.4).
180. Grease and oil lubrication not as specified (see 3.25.5).
181. Fungus- and moisture-resistance not as specified (see 3.26).
182. Composition not as specified (see 3.26.1).
183. Treatment and painting not as specified (see 3.27).

Source: https://assist.dla.mil -- Downloaded: 2026l0-22T21:10Z
Check the source to verify that this is the current version before use.

184. Paint color not as specified (see 3.27).
185. Camouflage pattern data not as specified (see 3.28).
186. Identification marking/labels not as specified (see 3.29).
187. Shipping data plates not as specified (see 3.29.1).
188. Instruction plates not as specified (see 3.29.2).
189. Warning plates not as specified (see 3.29.3).
190. Stenciling not as specified (see 3.29.4).
191. Workmanship not as specified (see 3.36).
192. Welding not as specified (see 3.36.1).
193. Welder or welding operator certifications not as specified (see 3..36.1.1).
194. Bolted connections not as specified (see 3.36.2).
195. Riveted connections not as specified (see 3.36.3).

4.5.2 Tests.

4.5.2.1 Test conditions. Prior to examination or test, the heater shall be lubricated with oils and greases specified in 3.25 and shall be fully charged with transfer oil as specified in 3.16. Lubricants shall be those designated for use in the ambient temperature at the place of test. Unless otherwise specified in a test, tests shall be performed without shelter and at the climatic conditions existing at the place of test. The contractor shall make provisions to dissipate heat output generated by the heater at rated capacity. The heater shall operate as specified herein without maintenance other than the contractor's recommended normal scheduled maintenance as established by a maintenance schedule prepared and submitted by the contractor prior to test.

4.5.2.2 Road.

4.5.2.2.1 Type II. The prime mover for these tests shall be a 5 ton series truck. Tow the heater for 25 miles over secondary roads at speeds necessary to average not less than 25 mph. Tow the heater for 25 miles over primary roads at speeds up to 45 mph, to average not less than 40 mph. In addition, tow the heater on smooth paved surfaces to include not less than 15 abrupt stops from a speed of 20 mph in the forward direction as specified in 3.18.1.2.2. Measure accurately the distance from initial movement of the brake pedal to stopping point for the 15 abrupt stops. Tow the heater over V-ditches not less than 36 inches wide and 12 inches deep. Observe ground contact areas for possible permanent deformation other than paint. Negotiate the ditches both perpendicular and at an angle of 45 degrees to the direction of travel, until at least 10 V-ditch crossings have been completed. Turn the prime mover and heater 60 degrees from straight ahead and perform the rolling turn in a circle for two complete turns and repeat in opposite direction. The suspension articulation shall be tested by traveling over 6-inch high obstacles which are square in the cross sections. Demonstrate compliance with DOT FMCSR for off-tracking of towed vehicles. Demonstrate compliance of the brake system with DOT requirements specified herein. Noncompliance with any DOT requirements loss of tire-to-ground contact of remaining wheels while one wheel negotiates the 6-inch high square obstacles, any operating difficulty; damage to or displacement of any component or accessory, installed on the heater; exceeding the specified stopping distance; evidence of' defective welded, bolted, or riveted connections; or evidence of any leakage from the heater, air lines, and hydraulic lines shall constitute failure of this test.

4.5.2.3 Heat exchanger. Operate the fully-charged heater until discharge flow rate and temperature are stabilized as specified in 3.8. Use a metering device

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:10Z
Check the source to verify that this is the current version before use.

which shall measure the flow of transfer oil within ±2 percent by weight of actual flow. Demonstrate the accuracy of the metering device by metering 100 gpm of transfer oil for a period of not less than 5 minutes. The oil shall flow through a closed circuit to the return port on the heater at which point the temperature of the transfer oil should be not more than 360 'F. Determine the pressure drop of transfer oil between inlet and outlet of the heat exchanger. Operate the heater for not less than 1 hour under these conditions and record the burner fuel consumption. Nonconformance to 3.8 c. and d. shall constitute failure of this test.

4.5.2.3.1 Pump. Operate the main circulating pump for 1 hour while pumping transfer oil at a temperature of 450°F and at a discharge pressure of not less than 50 psig. Nonconformance to 3.8 c. shall constitute failure of this test.

4.5.2.4 Heat exchanger insulation. Operate the fully-charged heater at rated output capacity for not less than 4 hours. Measure the temperature of the outer shell and rear head at not less than four locations on the outer shell and two locations on the rear head. Conduct this test during the last 4 hours of the endurance test specified in 4.5.2.6. A surface temperature at any location on the outer shell or rear head in excess of 125 'F, plus ambient temperature, or shield or guard provisions missing when the surface temperature at any location on the outer shell or rear head is in excess of 85 'F, plus ambient temperature, shall constitute failure of this test.

4.5.2.5 Cold-oil seal temperature. Operate the fully-charged heater at rated output capacity for not less than 8 hours. Measure the temperature of the transfer oil in the cold-oil seal tank or the expansion tank, when used as cold-oil seal, within 2 inches of the oil surface. A temperature at any point in excess of 150 'F, plus the ambient temperature, shall constitute failure of this test.

4.5.2.6 Endurance. Operate the fully-charged heater for not less than 104 hours at rated output capacity as specified in 3.8.a and 3.8.b. The transfer oil outlet temperature shall be at least 450 'F. Operate the heater continuously or for increments of not less than 8 hours. Upon completion of 104 hours of operation, open and examine the heat exchanger. Any evidence of cracking, flaking, shrinking, cement failure, or any other damage to the insulation material or any evidence of leakage, breakage, permanent deformation, or malfunction shall constitute failure of this test. Conformance to 3.8.a shall be determined either by calculations involving transfer oit kflow rate and heater inlet and outlet temperatures or by calculations performed on data colledted from material to which the transfer oil rejected heat. Nonconformance to 3.8 a. (see 6.2.2) and 3.8.b. shall constitute failure of this test.

4.5.2.7 Cycling. Start the fully-charged heater, operate for 1 hour, and shut down. Restart the heater and run through the complete operating sequence as specified in 3.13.1 for 25 times. Inability of the burner to sequence as specified in 3.13.1 or evidence of leakage or malfunction shall constitute failure of this test.

4.5.2.8 Expansion tank capacity. Heat the oil in the fully-charged system from 70 to 475 'F and measure the amount of oil in the expansion tank. Nonconformance to 3.16 shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:10Z
Check the source to verify that this is the current version before use.

4.5.2.9 Simulated rain. After the performance test has been completed, subject the heater to include the control box, all exposed safety components, electrical conduits, and junction boxes to a simulated rain described as follows for a period of 10 minutes: Four sprinkler-type nozzles with a 90 degree conical pattern shall be connected to a water supply system which will provide water to each nozzle at a nozzle pressure of not less than 50 psi. Nozzle capacities at the operating pressure shall be not less than 20 gpm. The four nozzles shall be spaced and directed at each longitudinal side of the heater so that the resulting spray pattern completely encompasses the heater. Any evidence of water inside the control box, exposed safety components, electrical conduits, or junction boxes of" the heater shall constitute failure of this test.

4.5.2.10 Transportability.

4.5.2.10.1 Tiedown provisions test. The tiedown provisions shall be tested in accordance with MIL-STD-209 to prove conformance to 3.23.1. Inability to meet the requirements of 3.23.1 shall constitute failure of this test.

4.5.2.10.2 Slinging provisions test. The slinging provisions shall be tested in accordance with MIL-STD-209 to prove conformance to 3.23.2. Inability to meet the requirements of 3.23.2 shall constitute failure of this test.

4.5.2.10.3 Air transportability test. The air transportability requirement for the C-130 and C-141 aircraft, in accordance with MIL-STD-1791 to prove conformance to 3.23.3. Inability to meet the requirements of 3.23.3 shall constitute failure of this test.

4.5.2.10.4 Rail transportability test. The heater shall be tested in accordance with MIL-STD-810 to prove conformance to 3.23.4. Inability to meet the requirements of 3.23.4 shall constitute failure of this test.

4.5.2.11 Control and safety device. Subject each control and safety device to functional operation for not less than 10 times each. Inability of any control or safety device to function as specified in 3.14 or evidence of leakage shall constitute failure of this test.

4.5.2.12 Electromagnetic interference. The heater shall be tested in accordance with SAE J551. A report tabulating the allowable maximum interference and the measured interference shall be furnished with the first article test report. Failure to meet the requirement of SAE J551 or failure to submit the required reports shall constitute failure of this test.

4.5.2.13 Noise level test. Noise levels shall remeasured in accordance with MIL-STD-1474 requirements and reported in the format indicated by MIL-STD-1474, figure 11. As a minimum, noise levels shall be measured when equipment is operating under full load. MIL-STD-1474, 5.1.2.1.4 contours shall be taken at not fewer than 12 equal (horizontal) arc increments, one increment shall include data from the noisiest position. Additionally, the noise level at the typical operating position shall be provided as dB(A) level. Failure to comply with MIL-STD-1474 provisions shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:10Z
Check the source to verify that this is the current version before use.

4.6 Inspection of packaging.

4.6.1 First article pack inspection.

4.6.1.1 Examination. The first article pack shall be examined for the defects specified in 4.6.2.3. Presence of one or more defects shall be cause for rejection.

4.6.1.2 Test. First article packaging shall be tested for its adequacy in accordance with the rail-impact test as set forth in MIL-STD-1186, appendix A.

4.6.1.2.1 Test failure criteria. Failure of the test shall be based on the failure criteria set forth in MIL-STD-1186, appendix A.

4.6.2 Quality conformance inspection of pack.

4.6.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.2.2 Sampling. Sample size shall be determined by using MIL-STD-105, table I and table IIa. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

4.6.2.3 Examination. Samples selected in accordance with 4.6.2.2 shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

No.	Defect	A	C
196.	Preservatives and their application not as specified.	5.2.1.1	
197.	Unprotected surfaces not preserved as specified.	5.2.1.2	5.2.2
198.	Trailer components not preserved as specified.	5.2.1.3	
199.	Openings in electric motors not sealed as specified.	5.2.1.4	5.2.2
101a.	Interior surfaces of the burner, fuel pump and lines not coated with preservative as required.	5.2.1.5	
102a.	Fuel line opening to fuel tank not sealed as specified.	5.2.1.5	5.2.2
103a.	Power cable not contained on reel as specified.	5.2.1.6	
104a.	Exposed cable ends not sealed as specified.	5.2.1.6	5.2.2
105a.	Flexible hose not positioned in container provided.	5.2.1.7	5.2.2
106a.	Hose container lid not secured as specified.	5.2.1.7	
107a.	Repair parts not preserved as specified.	5.2.1.8	5.2.2
108a.	Maintenance tools not preserved as specified.	5.2.1.9	
109a.	Technical publications not preserved as specified.	5.2.1.10	
110a.	Oil lubrication not specified.	5.2.1.11	
111a.	Exposed chains, when utilized, not preserved as specified.	5.2.1 .12.1	
112a.	Drivebelts and pulleys, when utilized, not preserved as specified.	5.2.1.12.2	
113a.	Enclosed gears and chains, when utilized, not preserved as specified.	5.2.1.12.3	
114a.	Flame safeguard relay, when utilized, not preserved as specified.	5.2.1 .12.4	
115a.	Consolidation not as specified.	5.2.1.13	5.2.2

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:10Z
Check the source to verify that this is the current version before use.

116a.	Level C preservation not as specified.		5.2.2
117a.	Skid-mounted heaters not packed as specified.	5.3.1.1	5.3.2
118a.	Trailer-mounted heater not prepared for shipment as specified.	5.3.1.2	5.3.2
119a.	Marking not as specified.	5.4	5.4

5. PACKAGING

5.1 First article pack. When specified (see 6.2), the contractor shall furnish a first article pack for examination and test within the time frame specified (see 6.2) to prove prior to starting production packaging that the applied preservation, packing, and marking comply with the packaging requirements of this specification. Examination and tests shall be those specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.6). The first article pack may be prepared utilizing either the first article heater or a production heater. When the first article model heater is utilized, the preservation, and packing shall be removed by the contractor at no expense to the Government, when requested by the Government, to facilitate comparison between the first article model heater and production heaters.

5.2 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Preservatives. Preservatives specified herein shall conform to the applicable specifications listed in and shall be applied in accordance with MIL-P-116.

5.2.1.2 Unprotected surfaces. Unpainted exterior metal surfaces of components not specifically specified herein and requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with type P-1 preservative.

.5.2.1.3 Trailer. Components of the trailer shall be preserved in accordance with the level A (mobile) preservation and packaging requirements of MIL-V-62038.

5.2.1.4 Electric motors. Openings into electric motors shall be sealed with tape conforming to PPP-T-60, type IV.

5.2.1.5 Burner. Interior surfaces of the burner, fuel pump, and lines shall be coated with P-10 preservative, grade 10. The fuel line opening to the fuel tank shall be sealed with a plastic or metal cap or plug, as applicable.

5.2.1.6 Power cable. The cable shall be contained on the reel and secured to prevent unreeling. The exposed end of the cable shall be sealed with tape conforming to PPP-T-60, type IV.

5.2.1.7 Flexible hose. The flexible hose shall be placed in the container. The container lid shall be secured in a manner to prevent pilferage of contents.

5.2.1.8 Repair parts. The preservative application criteria and applicable method(s) of preservation of MIL-P-116 shall be used to preserve repair parts.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:10Z
Check the source to verify that this is the current version before use.

5.2.1.9 Maintenance tools. Maintenance tools shall be preserved and packaged in accordance with PPP-P-40, level A.

5.2.1.10 Technical Publications. Technical publications shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3.

5.2.1.11 Oil lubrication. All lubricated components shall be filled to the operating level with winter grades of military oils designated for use in the temperature range of 0 to 32 'F. Each component shall be tagged in a conspicuous place to indicate the temperature range and grade of lubricant used.

5.2.1.12 Design features. When the following components are utilized in the design features of the burner, they shall be preserved as specified herein.

5.2.1.12.1 Exposed chains. Exposed drive chains shall be coated with enough type P-3 preservative to assure penetration of the preservative to the inner surfaces of the rollers, pins, and bushings. After the excess preservative has drained, the chains shall be coated with type P-1 preservative. Unpainted sprockets shall be coated with type P-1 preservative.

5.2.1.12.2 Drivebelts and pulleys. Belts shall be removed or released from tension. Removed belts shall be preserved in accordance with MIL-P-116, method It-l l Unpainted pulley grooves shall be coated with primer conforming to TT-P-664.

5.2.1.12.3 Enclosed gears and chains. Gears and chains enclosed within housings and with the housing filled with oil to the operating level as specified in 5.2.1.11 shall be actuated to assure complete coverage with the lubricant.

5.2.1.12.4 Flame safeguard relay. The flame safeguard relay shall be removed from the control box and preserved in accordance with MIL-P-116, method IC-1. The preserved relay shall be secured within the control box. A tag conforming to UU-T-81, type A, shall be attached to the control box indicating "Flame safeguard relay is disconnected and is secured within. Connect relay before operating heater."

5.2.1.13 Consolidation. The repair parts, maintenance tools, technical publications, ground rod assemblies, removed drivebelts when required, and loose components shall be consolidated in the toolbox when space permits. When space does not permit, the remainder of the items that will not fit in the toolbox shall be consolidated in a close-fitting box conforming to PPP-B-601, domestic type, style optional. The box shall be secured to the heater with strapping. Strapping shall conform to ASTM D 3953, type 1 or 2, zinc-coated, size as applicable and ASTM D 4675. The box shall be secured in a manner that will not increase cubage or interfere with lifting or towing of the heater. The toolbox lid shall be secured to prevent pilferage.

5.2.2 Level C. Each complete burner shall be preserved so as to prevent deterioration and damage during shipment to the initial destination and during short term storage. As a minimum, openings into electric motors, burner, and power cables shall be sealed with waterproof tape. Flexible hose shall be placed in its container. Repair parts shall be preserved as specified in 5.2.1.8. Consolidation shall be as specified in 5.2.1.13 except that the box for items not

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:10Z
Check the source to verify that this is the current version before use.

placed in the tool box shall conform to PPP-P-636, class weather-resistant, variety SW or DW, grade appropriate for the weight and size of contents.

5.3 Packing. Packing shall be level A or C as specified (see 6.2).

5.3.1 Level A.

5.3.1.1 Type I - skid-mounted heaters. Each skid-mounted heater, preserved as specified in 5.2, shall be packed in a crate conforming to MIL-C-104, type I, class 2, style a. Loading, bracing, blocking, bolting, cushioning and closure shall be in accordance with the appendix to the crate specification and with MIL-STD-1186.

5.3.1.2 Type II - trailer-mounted heaters. Each trailer-mounted heater, preserved as specified in 5.2, shall be prepared for unboxed. (mobile) shipment, The securing of the consolidation container to the trailer and the securing of the toolbox lid shall be as specified in 5.2.1.13. Wheels shall be free to rotate.

5.3.2 Level C. Complete type I heaters shall be packed in crates conforming to MIL-C-3774, type I, style a. Blocking, bracing, and tie-down shall be in accordance with MIL-STD-1186. Type II heaters shall be shipped mobile with the consolidation container secured to the trailer.

5.4 Marking. Marking for shipment and storage shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The heaters are intended to heat and circulate transfer oil for use with asphalt plants, asphalt melters, or other similar heating applications.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type and size required (see 1.2).
c* Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. Time frame required for submission of first article model, and number of heaters required (see 3.1).
e. When the Government will conduct any or all of the first article model examination and tests. When the Government will conduct some but not all of the first article examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.1).
f. When the identification of materials and finishes are required (see 3.4.3).
g. When transfer oil is not to be furnished (see 3.4.5).
h. When pipe and tubing will be other than specified (see 3.4.12).
i. When valves and pipe fittings will be other than as specified (see 3.4.13).
j. When the circulating pump motor will be design C (see 3.10).

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:10Z
Check the source to verify that this is the current version before use.

k. When main circulating pump shall be other than as specified (see 3.15.1).
l. When a 12-contact receptacle is not required (see 3.18.1. 13.1).
m. When a transportability report is required (see 3.23 and 6.2.1).
n. Color when other than specified (see 3.27).
o. When overcoating with camouflage patterns is required (see 3.27).
p* Time frame when camouflage pattern data is required (see 3.28).
q. When Government-furnished property is not required (see 3.30).
r. When the welding process shall not be at the option of the contractor (see .3.31.1).
s. When a first article pack is required, and time frame required for submission of first article pack (see 5.1).
t. Level of preservation and packing required (see 5.2 and 5.3).
u. Any special markings (see 5.4).

6.2.1 Consideration of data requirements. The following data requirements must be listed, as applicable, on the Contract Data Requirements List (DD Form 1423) when this specification is applied on a contract, in order to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Paragraph No.	DID No.	DID title	Suggested Tailoring
3.23	DI-PACK-80880	Transportability report	None
3.28	DI-MISC-80192	Color photograph prints	None
3.28	DI-MISC-80176	Camouflage line art data	None

The above DID's were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (ANSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.2.2 Certification. The contracting officer should arrange for delivery from the contractor certification for the following items:

Gaskets and seals (see 3.4.7).
b. Burner output (see 3.8.a).
Main circulating- pump (see 3.8.c).
d. Circulating pump motor (see 3.10).

6.3 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:10Z
Check the source to verify that this is the current version before use.

6.4 Lubricants. MIL-STD-838 prescribes the policy for using specification-type products wherever possible and provides specific requirements for potential use of non-standard proprietary products. MIL-STD-838 is implemented by MIL-HDBK-113. The contracting officer should note that unless otherwise authorized by the US Army Belvoir Research, Development, and Engineering Center (ATTN: STRBE-FL), Fort Belvoir, VA 22026-5606, lubricants, fluids, and greases for ground equipment systems must be restricted-to those listed under MIL-HDBK-113, chapter 2.

6.5 Government-furnished property. The contracting officer should arrange to . furnish the property listed in 3.30. - - · · ·

6.6 First article pack. Any changes or deviations of production packs from the approved first article pack will be subject to the approval of the contracting officer. Approval of the first article pack will not relieve the contractor of his obligation to preserve, package, pack, and mark the heaters in accordance with this specification.

6.8 Subject term (key word) listing.

Heaters, asphalt melters
Heaters, asphalt plant
Heaters, hot oil, trailer mounted, skid mounted

6.9 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ME

Preparing activity:
Army - ME

Review activity:
Army - AT

Project 3895-A334

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:10Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-H-52213F(ME)	2. DOCUMENT DATE (YYMMDD) 920629

3. DOCUMENT TITLE Heaters, Hot Oil

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (if applicable) (2) AUTOVON	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. NAME Betty Taylor	b. TELEPHONE (Include Area Code) (1) Commercial (703) 704-3466	(2) AUTOVON 654-3466
c. ADDRESS (Include Zip Code Army Belvoir RDE Center ATTN: STRBE-TSE Fort Belvoir, VA 22060-5606	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:10Z
Check the source to verify that this is the current version before use.